

IMPLEMENTATION SHOWCASE · 2026

Enterprise AI Knowledge Platform

An end-to-end document intelligence system — ingest, chunk, embed, search, compare, and converse with your organisation's knowledge using LLMs and RAG.

React + TypeScript

FastAPI · Python 3.11

MongoDB Atlas

Groq LLM

Vector + BM25 + Hybrid Search

LangGraph Agentic RAG

Docker · Nginx

Turborepo Monorepo

Stack

React · FastAPI · MongoDB
Atlas · Groq

Architecture

Monorepo — apps/frontend ·
apps/backend

Phases

6 phases + full test
pyramid

What the Platform Does

A full-stack internal tool that turns raw documents into a searchable, queryable knowledge base — supporting every step from ingestion to natural-language conversation.



Ingest

Upload files (PDF, DOCX, TXT, CSV), crawl URLs, or sync Confluence spaces. SSE progress streaming shows every stage.



Chunk & Embed

8 chunking strategies (fixed, recursive, semantic, structural...) visualised side-by-side. Sentence-transformer embeddings stored in Atlas.



Search

Vector (semantic), BM25 (keyword), and Hybrid search with configurable Top-K and reranking.



RAG Chat

Multi-turn conversation with cited sources. Standard RAG mode or LangGraph agentic mode for complex queries.



Compare

Diff two documents structurally (difflib unified diff) plus an LLM-generated semantic change summary.



Analytics & Admin

7-day KPI charts, search & RAG logs, user management, role-based access (admin / editor / viewer), audit trail.

TECHNOLOGY STACK

React 18 + TypeScript

Vite + Tailwind CSS

FastAPI · Python 3.11

MongoDB Atlas (Vector + BM25)

Groq — llama-3.1-8b / llama-3.3-70b

LangGraph Agentic RAG

sentence-transformers (all-MiniLM-L6-v2)

slowapi Rate Limiting

Docker Compose · Nginx

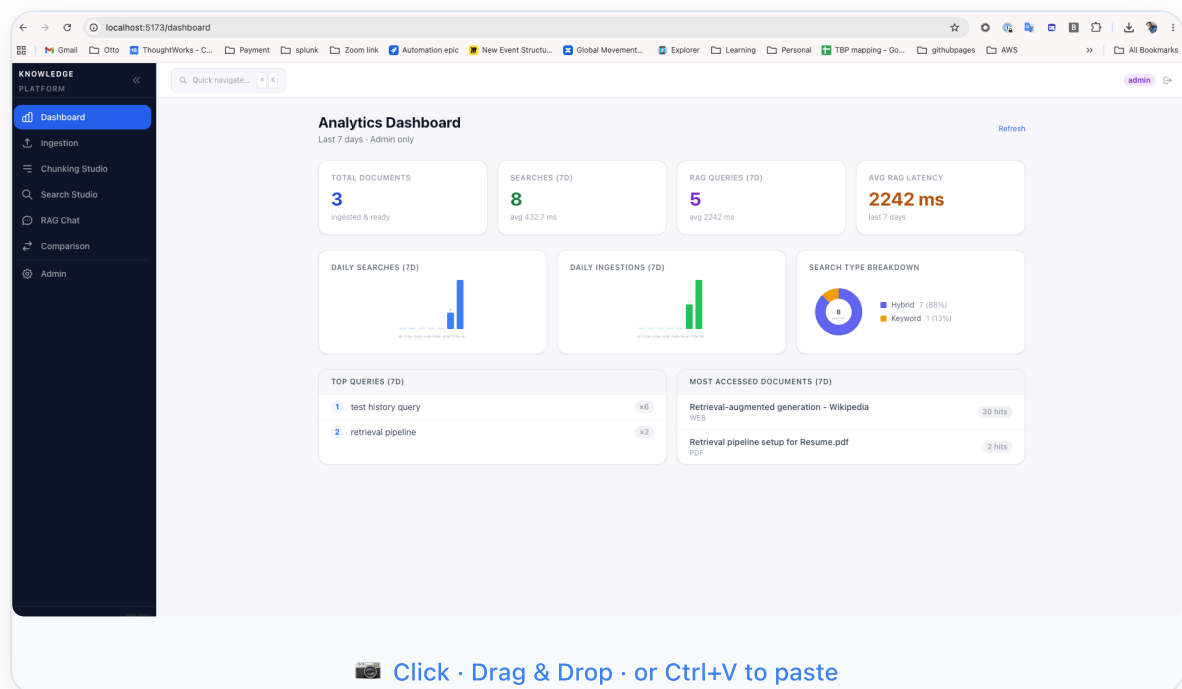
Turborepo Monorepo

Pytest · Vitest · Playwright E2E

Built across 6 structured phases — Core infra → Search → RAG → Agentic → Analytics
→ Production hardening — with a complete test pyramid: 28 backend unit tests,
integration tests with mongomock-motor, 19 Vitest component tests, and 20 Playwright
E2E tests.

Dashboard

The landing page after login. Live KPI cards and 7-day SVG bar charts give admins an at-a-glance view of platform activity — no chart library, pure SVG.



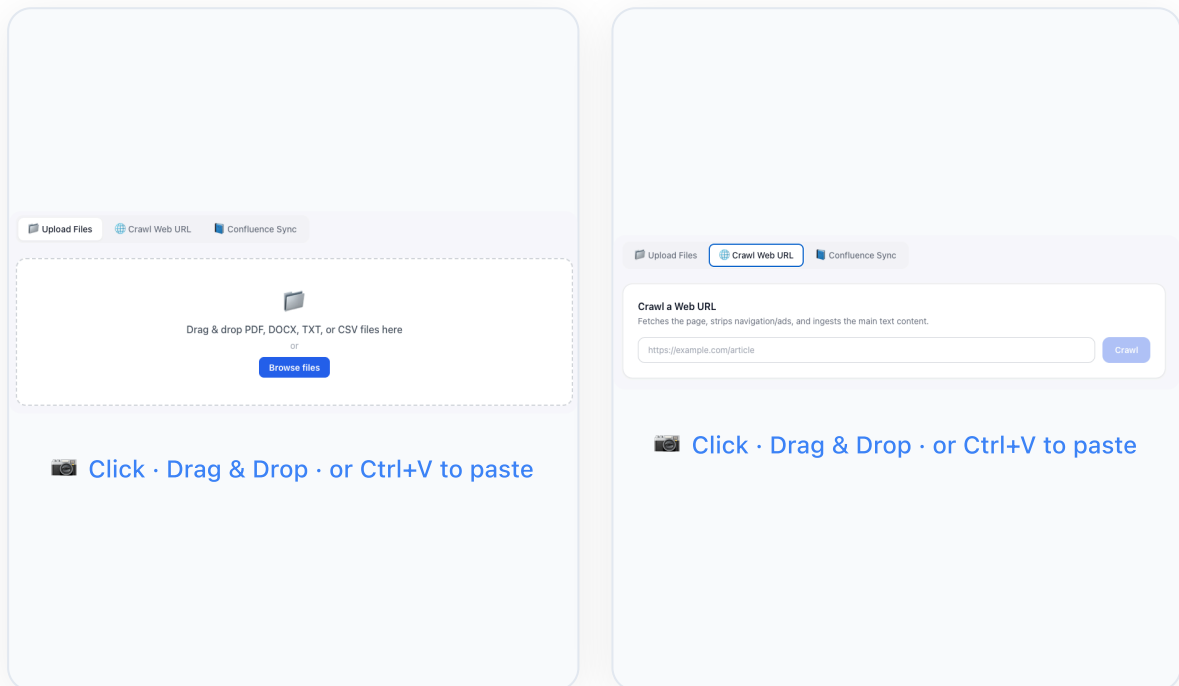
Dashboard — KPI cards and 7-day activity charts

DASHBOARD FEATURES

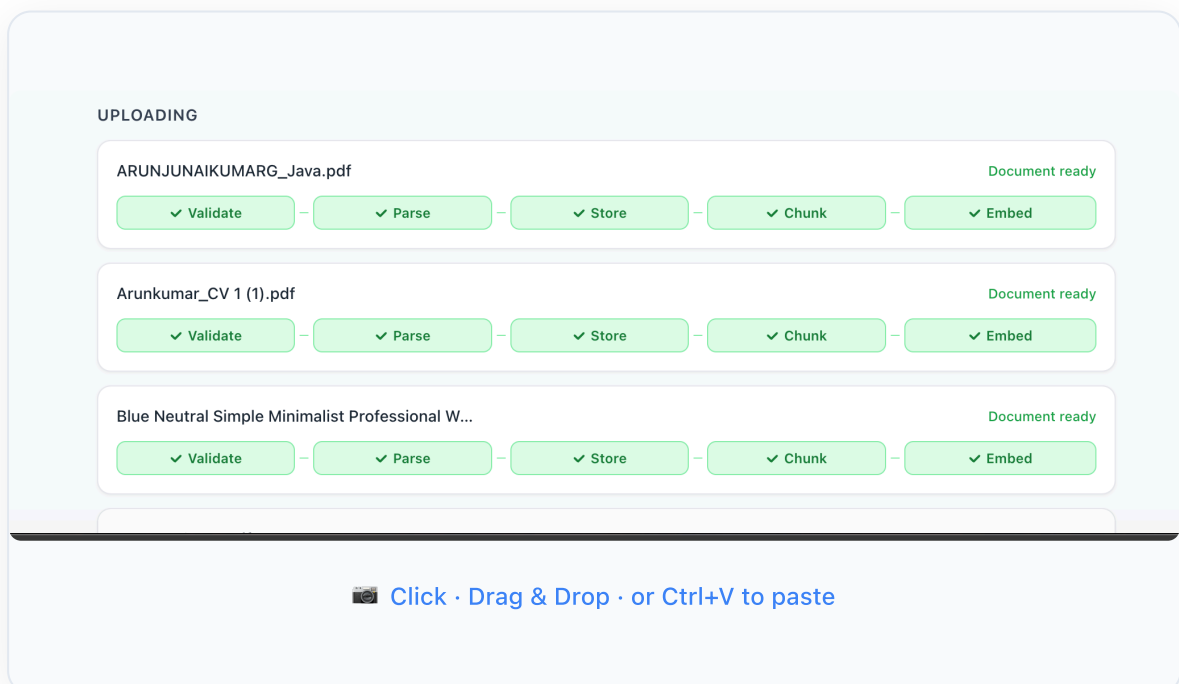
- ✓ Total documents KPI card
- ✓ Active users KPI card
- ✓ Daily search volume — SVG bar chart
- ✓ Search mode distribution — donut chart
- ✓ Admin-only access (RBAC gated)
- ✓ Total queries (7-day) KPI card
- ✓ Average response time KPI card
- ✓ Daily ingestion volume — SVG bar chart
- ✓ Top searched terms list
- ✓ Auto gap-fill for days with no activity

Ingestion

Upload files, crawl public URLs, or sync a Confluence space. Every ingestion streams real-time progress via Server-Sent Events — parse → chunk → embed → store.



File upload (left) and URL crawler (right)



Live SSE progress bar with stage-by-stage status

INGESTION FEATURES

- ✓ File upload (PDF, DOCX, TXT, CSV)
- ✓ Confluence space sync
- ✓ File size validation (configurable MB limit)
- ✓ 5-stage progress: Validate → Parse → Chunk → Embed → Store
- ✓ Content hash deduplication
- ✓ Public URL crawler (configurable depth)
- ✓ SSE real-time progress streaming
- ✓ MIME type + extension fallback validation
- ✓ Document library with delete (admin only)
- ✓ Ingestion audit trail logged

Chunking Studio

Select any ingested document and compare how 8 different chunking strategies split it. Click a chunk to highlight the corresponding source text — bidirectional.

The screenshot displays the 'Chunking Studio' interface. At the top, it says 'Side-by-side visualiser with 8 chunking strategies'. Below this, it shows '6 chunks', 'Avg 42 tokens', and 'Total 252 tokens'. The 'Document' dropdown is set to 'Arishraj Resume (1).pdf' and the 'Strategy' dropdown is set to 'LLM-Based (Grok)'. The interface is split into two main panels. The left panel, titled 'DOCUMENT · ARISHRAJ RESUME (1).PDF', contains the original text of the resume, which is color-coded by section: a blue header section, a green education section, a yellow work experience section, a purple skills section, and a pink contact/footer section. The right panel, titled 'CHUNKS · LLM_BASED', lists four chunks. Each chunk is color-coded to match its corresponding section in the document: #0 (blue), #1 (green), #2 (yellow), and #3 (purple). Each chunk entry shows the token count, character count, a preview of the text, and a 'strategy_detail: groq_llm' link. At the bottom of the interface, there is a button that says 'Click · Drag & Drop · or Ctrl+V to paste'.

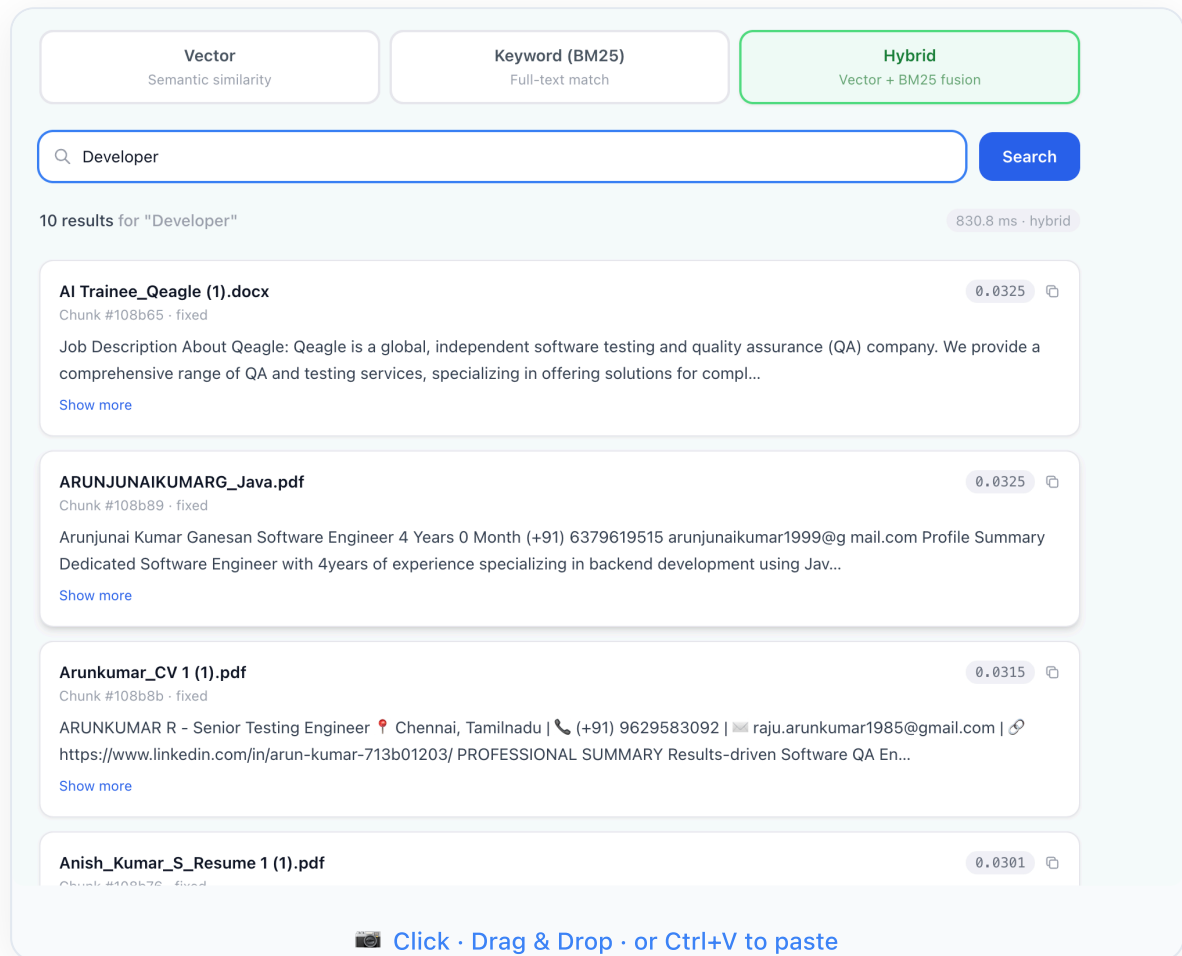
Side-by-side: original text (left) vs. chunk list (right) with click-to-highlight

CHUNKING STUDIO FEATURES

- ✓ 8 strategies: Fixed, Recursive, Semantic, Structural, Metadata-Aware, Query-Aware, LLM-Based, Agentic
- ✓ Click chunk → highlight source text
- ✓ Chunk count, avg size, overlap stats
- ✓ Lazy evaluation — computed on first request, cached
- ✓ Side-by-side split view
- ✓ Click source text → highlight chunk
- ✓ Strategy comparison dropdown
- ✓ Works on any ingested document

Search Studio

Compare Vector, BM25, and Hybrid search modes on any query. Results include source metadata, similarity scores, and a copy-to-clipboard button per card.



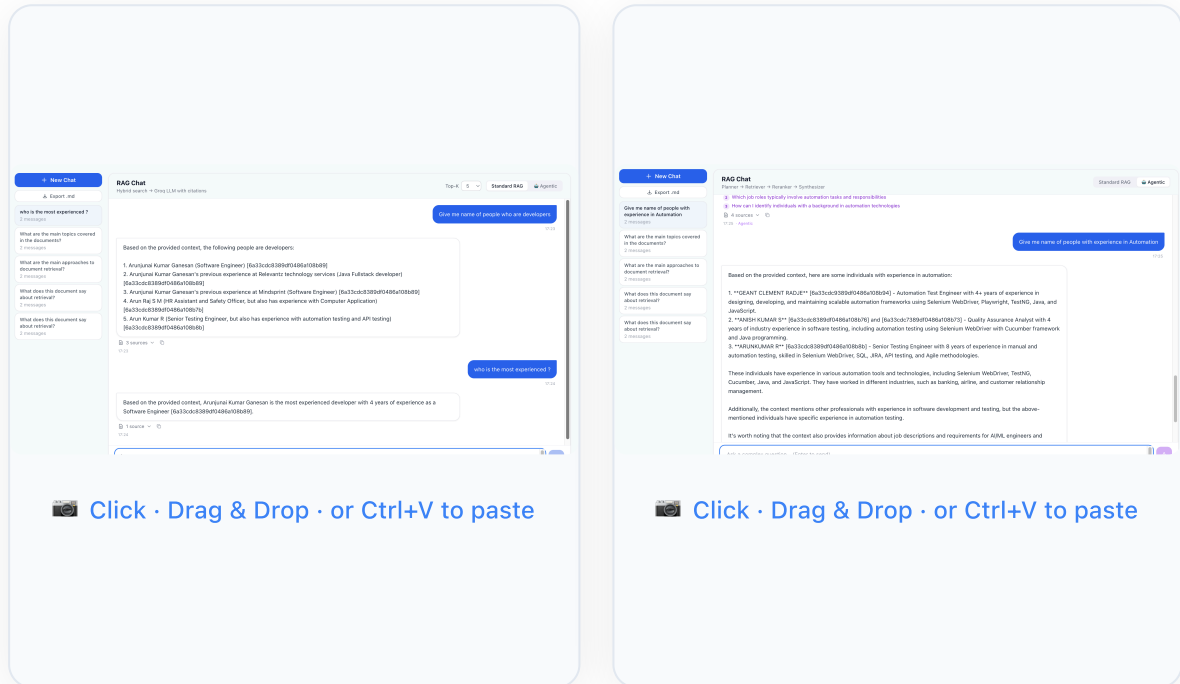
Search Studio — three search modes with ranked results and copy buttons

SEARCH STUDIO FEATURES

- ✓ Vector (semantic similarity) search
- ✓ Hybrid (vector + BM25 fusion) search
- ✓ Colour-coded search type buttons
- ✓ Source document metadata display
- ✓ Recent search history (localStorage, max 8)
- ✓ BM25 (keyword / full-text) search
- ✓ Configurable Top-K results
- ✓ Similarity score per result
- ✓ Copy-to-clipboard on each result
- ✓ Empty state for no results

RAG Chat

Multi-turn conversational interface over the knowledge base. Toggle between Standard RAG (fast, deterministic) and Agentic mode (LangGraph multi-step reasoning) per session.



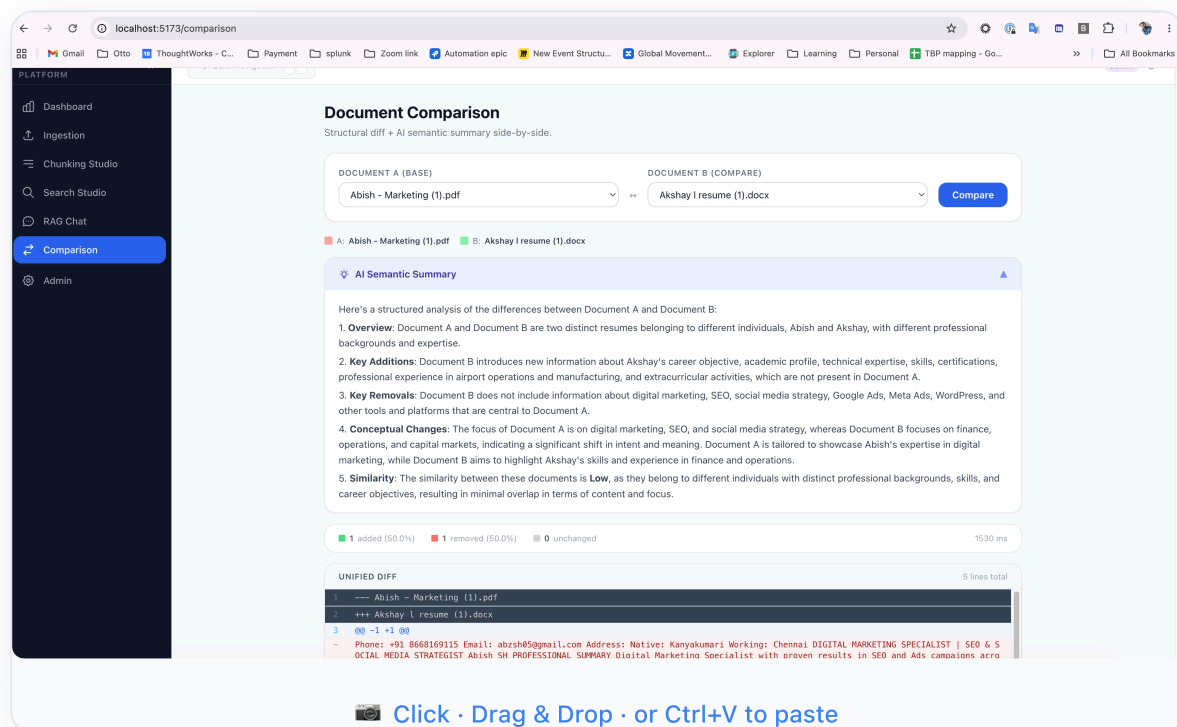
Standard RAG (left) and LangGraph Agentic mode (right)

RAG CHAT FEATURES

- ✓ Multi-turn conversation with memory
- ✓ Agentic mode (LangGraph + Llama-3.3-70b)
- ✓ Source citations with document links
- ✓ Session history sidebar
- ✓ Copy-to-clipboard per message
- ✓ New Chat clears session
- ✓ Standard RAG mode (Groq Llama-3.1-8b)
- ✓ Typewriter text animation (3 chars / 12ms)
- ✓ Top-K slider (standard mode)
- ✓ Export conversation to Markdown
- ✓ Suggestion chips on empty state
- ✓ Agentic mode badge + step indicators

Document Comparison

Select two documents to generate a unified diff view alongside an LLM-written semantic summary of what changed and why it matters.



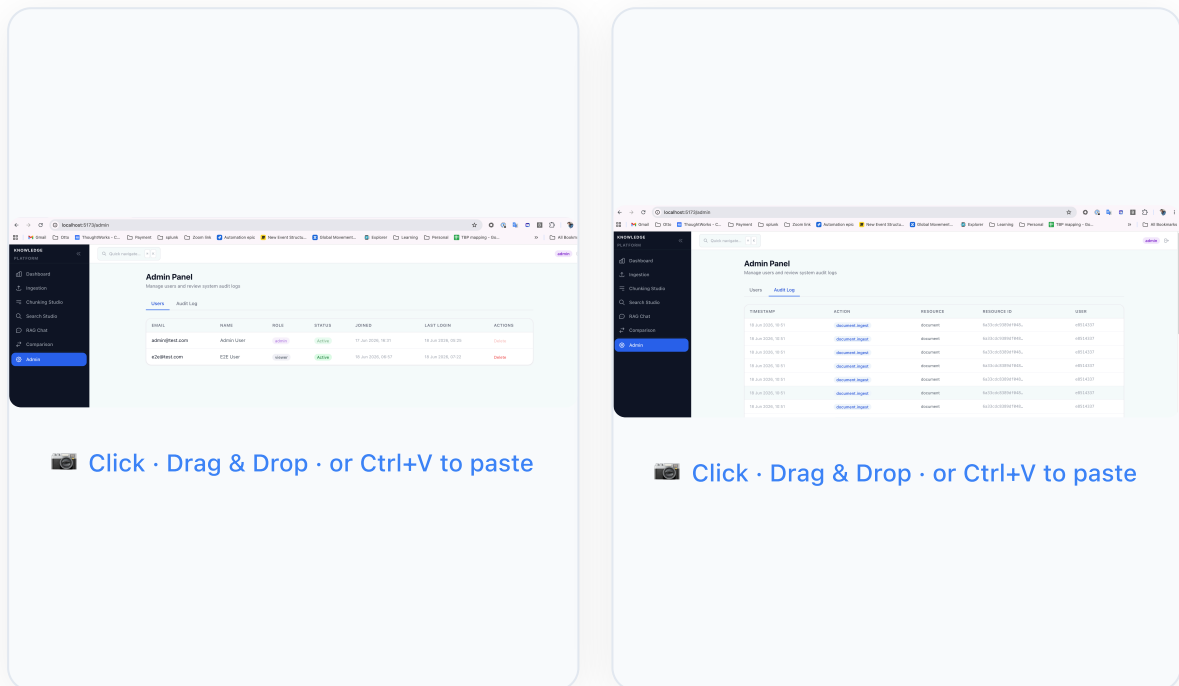
Unified diff panel (colour-coded) + AI semantic summary panel

COMPARISON FEATURES

- ✓ Select any two ingested documents
- ✓ Unified diff — added / removed / unchanged lines
- ✓ Colour-coded line-level diff (green / red)
- ✓ LLM semantic summary of changes
- ✓ Diff statistics (lines added, removed)
- ✓ Same-document guard (blocks self-comparison)
- ✓ Result cached by document-pair hash (order-independent)
- ✓ Admin + editor access

Admin Panel

Full user lifecycle management and a paginated audit log — all gated behind the admin role. Admins cannot modify their own account (self-edit guard).



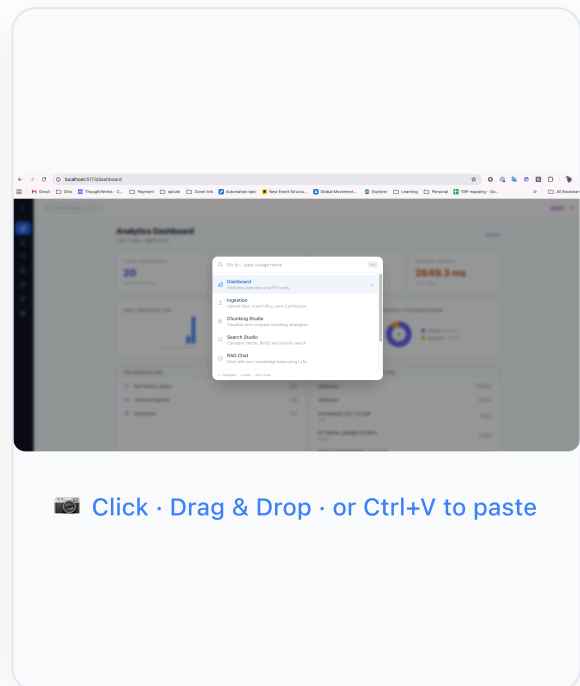
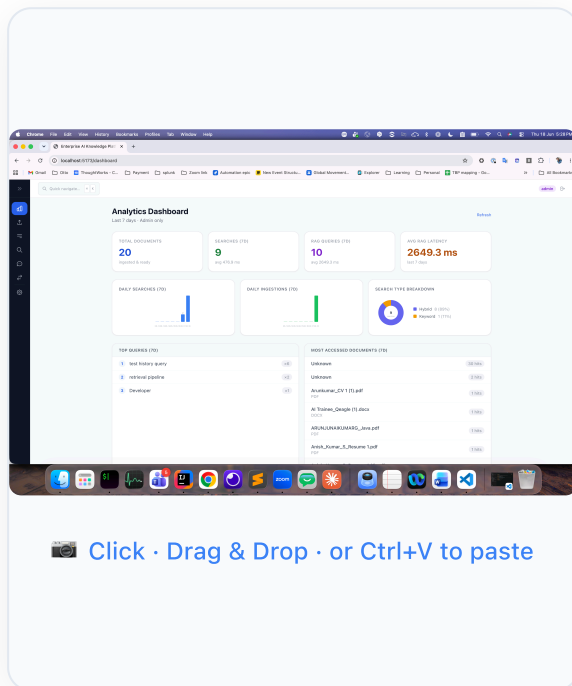
Users tab (left) and Audit Logs tab (right)

ADMIN PANEL FEATURES

- ✓ User list with role badges (admin / editor / viewer)
- ✓ Active / inactive toggle per user
- ✓ Self-edit guard (can't modify own account)
- ✓ Cursor-based "Load more" pagination
- ✓ Inline role change dropdown
- ✓ Delete user with confirm dialog
- ✓ Colour-coded action badges in audit log
- ✓ Admin-only RBAC gate (403 for non-admins)

Shared UI Components & Shell

A consistent shell wraps every page — collapsible sidebar, command palette, toast system, and a role-aware topbar — making the app feel like a polished product.



Collapsible sidebar with SVG icons (left) · ⌘K command palette (right)

UI SYSTEM FEATURES

- ✓ Collapsible sidebar (state persisted in localStorage)
- ✓ ⌘K / Ctrl+K command palette with fuzzy filter
- ✓ Global toast system (success / error / warning / info)
- ✓ Topbar: ⌘K shortcut button + role badge + sign-out
- ✓ Typewriter text effect on RAG responses
- ✓ CopyButton with "Copied!" confirmation state
- ✓ SVG Heroicons throughout (no icon library)
- ✓ ↑↓ keyboard navigation in palette
- ✓ Toast portal-rendered, slide-in CSS animation
- ✓ Role badge colours: purple=admin, blue=editor, grey=viewer
- ✓ EmptyState component with CTA action
- ✓ Loading spinner with brand colour

Test Pyramid

67 tests across four layers — backend unit, integration (mongomock-motor), frontend Vitest component tests, and Playwright E2E — all passing.



Backend Unit — 28 tests

Pure function tests:
_fill_daily(),
_parse_unified_diff(),
MIME validation, file size,
cache key symmetry. Zero I/O.



Backend Integration

mongomock-motor in-memory DB. Auth routes, document CRUD, analytics RBAC gates tested end-to-end through FastAPI.



Frontend Components — 19 tests

Vitest + Testing Library.
CopyButton, EmptyState, CommandPalette — keyboard nav, filtering, click handlers.

Q Search tests

All 20 ✓ Passed 20 Failed 0 Flaky 0 Skipped 0 🔄 ⚙️

18/06/2026, 12:52:18 Total time: 8.5s

> auth.spec.ts

> rag-chat.spec.ts

> search.spec.ts

🖱️ Click · Drag & Drop · or Ctrl+V to paste

Playwright E2E — 20 tests: auth flows, search studio, RAG chat

E2E TEST COVERAGE

✓ Register → login → dashboard flow

✓ Wrong password stays on login (security)

- ✓ Unauthenticated redirect to login
- ✓ Search type toggle buttons
- ✓ Recent search history dropdown
- ✓ RAG suggestion chips fill input
- ✓ Top-K visible / hidden per mode

- ✓ Sign-out redirects to login
- ✓ Search button enable/disable state
- ✓ ⌘K palette open and close
- ✓ Mode toggle Standard ↔ Agentic
- ✓ storageState — login runs once per suite

Docker & Deployment

One command to run the full stack in production mode. Two-stage Docker builds, Nginx reverse proxy, HuggingFace model volume, and rate-limited endpoints.

Start everything: `cd docker && docker compose up --build`
Frontend → <http://localhost> · API docs → <http://localhost/docs>



Backend Dockerfile

Two-stage: builder installs deps into `/opt/venv`, runtime copies it. Non-root app user. Healthcheck on `/health`.



Frontend Dockerfile

Node 18 Alpine builder → Nginx 1.27 Alpine runtime. 1-year immutable cache for hashed assets, no-cache for `index.html`.



Nginx Reverse Proxy

Routes `/api` → `backend:8000`.
`proxy_buffering off` for SSE streaming. SPA fallback for React Router.

PRODUCTION FEATURES

- ✓ Two-stage Docker builds (minimal runtime images)
- ✓ SSE streaming through proxy (buffering disabled)
- ✓ Rate limiting: register 5/min, login 10/min
- ✓ Non-root container user
- ✓ All secrets via `.env` (never in image)
- ✓ Nginx reverse proxy (single port: 80)
- ✓ HuggingFace model cached in named volume
- ✓ CORS configured via `.env`
- ✓ Health check endpoint `/health`
- ✓ Docker Compose orchestration